



Sotaro Takeshita

PHD CANDIDATE

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Education

University of Mannheim

PH.D. STUDENT IN NATURAL LANGUAGE PROCESSING

- Text summarization for scholarly documents.
- Supervisor: Prof. Dr. Simone Paolo Ponzetto

Mannheim, Germany

Sep. 2022 - Present

The National University of Electro-Communication

M.S. IN COMPUTER SCIENCE

- Thesis: Multimodal models for commonsense reasoning.
- Supervisor: Prof. Dr. Yasuhiro Minami

Tokyo, Japan

Apr. 2018 - Mar. 2020

The National University of Electro-Communication

B.S. IN COMPUTER SCIENCE

- Thesis: Unsupervised text summarization with reinforcement learning.
- Supervisor: Prof. Dr. Yasuhiro Minami

Tokyo, Japan

Apr. 2013 - Mar. 2018

Publication

JOURNAL

Cross-lingual extreme summarization of scholarly documents

SOTARO TAKESHITA, TOMMASO GREEN, NIKLAS FRIEDRICH, KAI ECKERT AND SIMONE PAOLO PONZETTO

International Journal on Digital Libraries (IJDL)

2023

CONFERENCE

IRSum: One Model to Rule Summarization and Retrieval

SOTARO TAKESHITA, SIMONE PAOLO PONZETTO, KAI ECKERT

To appear at GEM Workshop at ACL 2025

2025

GenGO Ultra: an LLM-powered ACL Paper Explorer

SOTARO TAKESHITA, SIMONE PAOLO PONZETTO, KAI ECKERT

To appear at The 63rd Annual Meeting of the Association for Computational Linguistics, System Demonstrations

2025

GenGO: ACL Paper Explorer with Semantic Features

SOTARO TAKESHITA, SIMONE PAOLO PONZETTO, KAI ECKERT

The 62nd Annual Meeting of the Association for Computational Linguistics, System Demonstrations

Bangkok, Thailand

2024

ROUGE-K: Do Your Summaries Have Keywords?

SOTARO TAKESHITA, SIMONE PAOLO PONZETTO, KAI ECKERT

The 13th Joint Conference on Lexical and Computational Semantics colocated at NAACL 2024, (Best Long Paper)

Mexico City, Mexico

2024

ACLSum: A New Dataset for Aspect-based Summarization of Scientific Publications

SOTARO TAKESHITA, TOMMASO GREEN, INES REINIG, KAI ECKERT AND SIMONE PAOLO PONZETTO

In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)

Mexico City, Mexico

2024

ZusammenQA: Data Augmentation with Specialized Models for Cross-lingual Open-retrieval Question Answering System

CHIA-CHIEN HUNG, TOMMASO GREEN, ROBERT LITSCHKO, TORNIKE TSERETELI, SOTARO TAKESHITA, MARCO BOMBIERI, GORAN GLAVAŠ, SIMONE PAOLO PONZETTO

Workshop on Multilingual Information Access (MIA) at NAACL

Seattle, Washington + Online

2022

X-SCITLDR: Cross-Lingual Extreme Summarization of Scholarly Documents

SOTARO TAKESHITA, TOMMASO GREEN, NIKLAS FRIEDRICH, KAI ECKERT AND SIMONE PAOLO PONZETTO

ACM/IEEE Joint Conference on Digital Libraries (JCDL)

Cologne, Germany and Online

2022

Randomly Removing 50% of Dimensions in Text Embeddings has Minimal Impact on Retrieval and Classification Tasks

Suzhou, China

SOTARO TAKESHITA, YURINA TAKESHITA, DANIEL RUFFINELLI, SIMONE PAOLO PONZETTO

2025

Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (oral, EMNLP), **People's Choice Award**

Experience

Amazon

Luxembourg

APPLIED SCIENTIST, INTERNSHIP

Nov. 2025 -

- I worked on a project focused on evaluating the robustness of LLM-based web agents against injected UX defects. We developed a framework that dynamically modifies HTML elements while an agent navigates web pages, enabling the creation of two datasets of agent trajectories containing intentionally introduced UX issues (e.g., a link redirecting to an unexpected page). Using these datasets, we analyzed the performance and behavior of eight state-of-the-art visual large language models in detecting such defects. The resulting paper presenting our findings is currently under review for a conference.

VADIS (VARIABLE Detection, Interlinking and Summarization)

Germany

RESEARCHER

2022 - Present

- A project funded by German Research Foundation (DFG)
- Development of a summarization system for social science research articles.

Build It Inc.

Tokyo, Japan

DATA ANALYST & ML ENGINEER

2015 - 2022

- Analysis of user behaviours using statistical and machine learning tools
- Development of recommendation SaaS

The University of Campinas, Faculty of Electrical and Computer Engineering

Campinas, Brazil

RESEARCH INTERNSHIP

Nov. 2018

- Developed Neural Network-based Computer Vision system for self-driving wheelchairs.
- Advisor: Prof. Dr. Eric Rohmer

NTT Media Intelligence Lab

Kanagawa, Japan

RESEARCH INTERNSHIP

Sep. 2018

- Development of an NLP system to automatically generate funny stories in Japanese.
- Advisor: Dr. Ryuichiro Higashinaka

Recruit Holdings

Tokyo, Japan

DATA SCIENCE INTERNSHIP

Aug. 2017

- Development of an automatic scoring system for essays.
- Prized with a best intern award

Sección de Estudios de Posgrado e Investigación de ESIME Culhuacan, Instituto Politecnico Nacional

Mexico City, Mexico

RESEARCH INTERNSHIP

Feb. 2016 - Apr. 2016

- Analysis on models of automatic face recognition models.
- Advisor: Prof. Dr. Mariko Nakano Miyatake

Projects

GenGO: ACL Paper Explorer with Semantic Features, <https://gengo.sotaro.io>

GenGO-Chat: LLM-powered NLP paper exploration system., <https://gengo-chat.sotaro.io/>

Talks

2024 **Scholarly Document Processing: Language Technologies to Help Scientist**, Invited Talk

DESY, Germany

2023 **Overview of Tasks and Challenges in Automatic Text Summarization**, Invited Talk

NEC, Germany